PAPER_951: Cooper Pair Phonon Coupling

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** bcs_superconductivity_uqff.py (CooperPairPhononCoupling) **Calculator:** CooperPairPhononCouplingCalc (CP4 #535) **CVW:** v2.0.0 compliant

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Abstract

We derive the effective Cooper-pair coupling strength via SCm phonon exchange at 1.25 THz. The coupling $V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \Phi_{1.25\,\text{THz}}(\omega, \Gamma)$ is modulated by the phonon resonance profile $\Phi = \exp(-(\omega - \omega_{\text{SCm}})^2/(2\Gamma^2)) \cdot S_{26}$, yielding pair binding energy $E_{\text{pair}} = 2\Delta(T)$.

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1. Coupling Strength

$$V_{\text{eff}}(\omega, \Gamma) = V_{\text{SCm}} \cdot \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26}$$

2. Pair Binding Energy

$$E_{\text{pair}} = 2\Delta(T)$$

At on-resonance ($\omega = \omega_{\text{SCm}}$), $\Phi = S_{26}$ and coupling is maximized.

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3. Source Data

- **File:** bcs_superconductivity_uqff.py
- **Session:** 214
- **CP4 Class:** CooperPairPhononCouplingCalc (#535)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Cooper, L.N. (1956) – Phys. Rev. 104, 1189